

CONTINUATION — HelixTranslate

session-resumption file

Revision: 18 **Last modified:** 2026-06-13T00:00:00Z **Purpose:** Single canonical out-of-the-box entry point for any fresh session (§11.4.131 / §12.10 / §11.4.127). To resume: point a new session at THIS file, run `git fetch --all`, and say **continue**.

Session wrap-up — W13, W15 (api/grpc), W16 all resolved (HEAD 1cc66ac)

- **W15 api/grpc DONE (real, evidence-backed)** — subagent-driven. Built build-tagged integration suites: `pkg/api/server_realhttp_integration_test.go` (real HTTP via `httptest` + real JWT + real Postgres via `brokertest`: `health`, `JWT 401/200`, `login→token` round-trip, / `api/v1/verified-models`, `session persist` round-trip) and `pkg/grpc/server_storage_integration_test.go` (real gRPC over `bufconn` + real Postgres: `submit→persist`, `status`, `cancel`, `list-count`, `8× concurrent`, `error-code` via `status.FromError`; §1.1 mutation-proven). LLM-translate leg = honest §11.4.3 SKIP (no provider key). Normal `go test` boots NOTHING; SKIP-clean without `podman`.
- **REAL AUTH BUG fixed (W15 api leg, commit 0e2a999)** — `InMemoryUserRepository.Create` stored the caller's pointer; `CreateUser` clears `Password=""` post-Create → blanked the stored bcrypt hash → EVERY user registered via `CreateUser` could never log in. Fix: repo stores a COPY. Conductor-reproduced RED(401)→GREEN(200) on `TestAPIRealHTTP_LoginTokenRoundTrip` (§11.4.115). Remaining W15: only the pure LLM-translation E2E (operator-gated on a real provider key).
- **W13 DONE (commit 1cc66ac)** — 8 PascalCase `.gitmodules` section labels → lowercase (paths were already compliant). Done §9.2 backup-first; all 11 submodules verified resolving to identical baseline SHAs after `git submodule sync`, dirty challenges/helix_qa preserved, build exit 0, backup removed post-gate.
- **W16 RESOLVED = KEEP Models/ as-is** — the `pkg/modelsbridge` bridge already consumes `digital.vasic.models` (functional need met). Converting `Models/` → a git submodule requires creating a NEW upstream repo under an owned org — an operator-gated repo-creation action (§11.4.101 block-only-when-irreversible+undeterminable; §11.4.122 keeping is the safe no-change path). Convert path remains available on operator request.

Autonomous-safe queue now EMPTY. Genuinely-remaining = operator-gated only: W15 LLM-translation E2E (real provider key) + W16 convert (new upstream repo decision). Per §11.4.94 this is legitimate idle.

LLMsVerifier real-key integration — DONE (HEAD e7d6d67; submodule 95fb1cce)

Operator directive: use `~/api_keys.sh` so the LLMsVerifier submodule performs its mandatory steps and provides valid providers + models to the System. **Done, evidence-backed (§11.4.83 docs/qa/llmsverifier_20260613_182600/, key-free):** - Ran the submodule's REAL discovery/verification with real keys → **89 verified models** (DeepSeek 2, Cerebras 2, Groq 16, Mistral 69; OpenAI/Anthropic honestly fail — no key in `api_keys.sh`). - **Two real bugs found + fixed, both §11.4.135 mutation-proven guards (exact production errors):** - **Bug B (main, internal/verifier/client.go) load-bearing:** the System's SSOT client couldn't decode the server's `{"count":N,"models":[...]}` envelope → consumed ZERO models while server healthy. Fixed (envelope-aware decode). Guard: `client_envelope_test.go`. System now fetches all 89 e2e. - **Bug A (submodule llmverifier/models.go, 95fb1cce, pushed GitHub+GitLab):** numeric `context_window` (Groq) crashed the whole provider decode → 0 models. Fixed (tolerant `UnmarshalJSON`). Guard: `context_window_unmarshal_test.go`. - Secrets: keys sourced inline only, never printed/logged/committed (§11.4.10); evidence dir + edits secret-scanned clean; throwaway run artifacts removed (§11.4.14).

Honest follow-up gaps (tracked, not bluffed — §11.4.6): - (G1) Submodule `verify` (report) pipeline does NOT persist into the server SQLite DB — the served DB was seeded via live discovery (verifier's own client+database pkgs). Architectural bridge worth a workable item. - (G2) Full ~30-provider exhaustive per-model coding-capability scoring timed out at 600s in one pass; reliability-scored the seeded set. A bounded/batched sweep is the follow-up. - (G3) Providers without keys (OpenAI/Anthropic/Perplexity/Together) fail auth honestly; add their keys to `api_keys.sh` to include them.

Real-bug-hunt waves (HEAD eb7576e) — 17 real bugs fixed this session, all reproduce-first + mutation-proven + pushed

LLMsVerifier real-key integration: 89 verified models flow to the System end-to-end; System's own in-process verifier env-bridge (31 models, `cmd/verify-models`); real DeepSeek translation proof ("The sky is blue today."->"El cielo esta azul hoy.").

Bugs fixed (each RED->GREEN, §11.4.135 guard, §1.1 mutation-proven, real evidence; commits `e7d6d67/dfb250c/f80153b/eb7576e` + submodule `95fb1cce`): - Auth lockout (`CreateUser` blanked stored bcrypt hash); Verifier consume (`client.go` envelope); Verifier serve (`listVerifiedModels` dropped all); Groq `context_window` (submodule); Hardcoded RU->SR prompt (RU->SR Ekavica preserved); Uppercase Cyrillic digraph (LJUBAV); Ordered-list numbering lost; UTF-8 mid-rune truncation (HIGH, Cyrillic); Import data race; FB2 inline+tail text dropped (CRITICAL); FB2 writer escaped markup; FB2 format misdetection; EPUB OPF/NCX id mismatch; Markdown inline whitespace collapse; silent chapter drop; list reverse round-trip; `extractJSON` wrong span.

Remaining tracked follow-ups (honest, §11.4.6/§11.4.118): - G1: submodule `verify`-pipeline->server-DB bridge (serial submodule wave, §11.4.119 contends with main builds; conductor-owned). - G2: batched full ~30-provider verification sweep (one-pass timed out 600s). - G3: add OPENAI/ANTHROPIC/etc keys to `api_keys.sh` (operator) to include those providers. - Deferred-broader: `markdownToHTML`

round-trip edges; preparation extractJSON LLM-output spec; verification multipass duplicate DB writes; ambiguous nj/lj/dz Latin->Cyrillic (structural §11.4.112 won't-fix).

Bug-hunt wave 3 (HEAD 6c76c3f) — 6 more real bugs (session total ~23), all mutation-proven + pushed

- CRITICAL OpenAIClient panic on non-float64 temperature (blast radius = ALL OpenAI-compatible providers) — toFloat64/toInt coercion.
- CLI -temperature/-max-tokens ignored by openai/anthropic/qwen/zhipeu — Options>typed-field>default precedence.
- Gemini discarded valid MAX_TOKENS partial text (data-loss) — accept STOP/MAX_TOKENS/"" (gemini_test.go reconciled §11.4.120).
- EventBus.Publish unbounded goroutine-per-handler + RLock-held-across-spawn → explosion + writer-starvation livelock — synchronous dispatch, lock released first (-race clean).
- LLMInstance.Available/LastUsed data race (mutex in 1/5 sites) — accessors (-race clean).
- APIKeyStore unsynchronized map → fatal concurrent-map panic (DoS) — RWMutex (-race clean). JWT/rate-limit confirmed already-hardened.

NEW tracked finding (this wave)

- **pkg/translator/llm full go test HANGS when api_keys.sh is in-env** — a NON-tagged test reaches the network (§11.4.98 violation: unit tests must be deterministic/offline). The wave's new defect tests are httptest/deterministic; the hang is in a pre-existing test. FIX: build-tag or httptest-ize the offending test. (Bounded go test -run <name> works fine.)

Remaining queue (for fresh-session continuation per §11.4.141/§11.4.127)

- G1: submodule verify-pipeline->server-DB bridge (serial submodule wave; §11.4.119 contends with main builds — conductor-owned, NOT parallel with main-repo subagents).
- G2: batched full ~30-provider verification sweep (one-pass timed out 600s).
- NET: fix the non-tagged network-reaching test in pkg/translator/llm (§11.4.98).
- G3: add OPENAI/ANTHROPIC/etc keys to api_keys.sh (operator) to widen provider coverage.
- More untouched packages to bug-hunt: pkg/storage (deeper), pkg/progress, pkg/report, pkg/logger, cmd/* entrypoints, pkg/grpc (the flagged send-on-closed-channel at server.go:403 — events pkg has no Unsubscribe; real follow-up).
- Deferred-broader: markdownToHTML round-trip edges; gemini GenerationConfig hardcoded (same class as fix #2); verification multipass duplicate DB writes.

SHORT resumption sentence

Read docs/CONTINUATION.md then .remember/remember.md, run git fetch --all, and **continue** the HelixTranslate “raise-to-enterprise” mandate at **Phase 4 (tests/Challenges/HelixQA to ~100% per type + expand real ebook-translation evidence)** while finishing the Phase-3 tail (Models decision, governance gaps). Binding: anti-bluff §11.4 (real captured evidence, no false PASS), no-force-push §11.4.113, flat snake_case submodules §11.4.28/§11.4.29, every change independently reviewed §11.4.142, containers-first infra §11.4.76, release tags prefixed helix_translate- (§11.4.151). Work subagent-driven (§11.4.70) with ≥3 parallel streams (§11.4.103).

Live state anchors (moment-valid)

- **Parent HEAD:** see git log (W14b adoption commit via commit_all.sh) on main; pushed to BOTH milos85vasic/Translator + HelixDevelopment/HelixTranslate (verified, fast-forward, no force). Session commit chain: b3dc7f9(llm_provider) → e775088 → 8a6af67 → 7331c54 → 76677aa → 18c5137 → de256dd → 9004477 → 13d5e30 → af1ef47 → a9b38fa → 218bfa0 → 1f06bf5 → a0930fc → 3027da2 → b2377af → 6d4ae47.
- **llm_provider HEAD:** b3dc7f9 (W7 CONST-036 propagation block + regenerated html/pdf), pushed to HelixDevelopment/LLMProvider master.
- **Constitution HEAD:** 5e671fe (§11.4.151 added), pushed to all 6 upstreams; parent pointer bumped.
- **Build:** go build ./... = EXIT 0. **Total test coverage = 50.7%** (go tool cover -func total; see docs/testing/coverage_matrix.md).
- **Submodules (flat, snake_case):** challenges containers constitution doc_processor docs_chain helix_qa llm_orchestrator llm_provider llms_verifier security vision_engine.
- **Release prefix:** HELIX_RELEASE_PREFIX=helix_translate (.env git-ignored, .env.example tracked). First RC: helix_translate-1.0.0-dev-0.0.1 — create ONLY when genuinely green (§11.4.40), operator-gated.
- **Safety backup:** ../helix_translate_gitbackup_20260611T181545Z/repo.git.mirror (pre-rename 311c585).

DONE

Phase 1 — flat snake_case submodules (reviewed GO, pushed): 8 submodules renamed; Go pkg challenges/ → pkg/challenge_runner/; docs_chain added; .env release-prefix. go build ./... EXIT 0. **Phase 2 — constitution §11.4.151** release-prefix rule → pushed to all 6 constitution upstreams; parent pointer bumped; inheritance meta-test PASS. **Phase 3 (partial):** - Security/ plain dir → flat security submodule (proven byte-identical to upstream, zero loss; §11.4.124). - All submodules + main synced to every upstream (one stale containers gitlab mirror FF'd). - **docs_chain OPERATIONAL:** .docs_chain/contexts/ tracked_docs.yaml registered; real pandoc+weasyprint sync produced docs/CONTINUATION.{html,pdf} + README.{html,pdf}; verify = in-sync. See docs/DOCS_CHAIN.md. - **Real translation PROVEN (anti-bluff):** EN → Serbian via DeepSeek deepseek-chat, 208 Cyrillic chars, differs from source, no placeholders. Evidence: qa-results/translation/<ts>/ (raw, git-ignored). Asset: test/

assets/crow_and_pitcher_en.txt. - **Fixed a regression I introduced:**
challenge scripts' PROJECT_ROOT="\${SCRIPT_DIR}/../../.." broke when moved 1
level deeper → corrected to ../../..; 3 challenges re-verified PASS
(cache_invalidation, model_verification_gate, anti_bluff step1-3).

CLOSED this session (2026-06-13, batch e775088 / llm_provider b3dc7f9 — all pushed, captured evidence)

- **W4**  pkg/security flaky Short-TTL — widened TTL→1min; 50× + -race + CPU-load all PASS.
- **W5**  pkg/distributed 120s timeout — bounded SMTP send (real defect) + in-process test servers; 3× ~48.5s, vet 0.
- **W6**  internal/verifier/discovery 600s timeout — injectable URLs (prod unchanged) + sentinel stub + honest SKIP; 3× ~1s, paired §1.1 mutation proven.
- **W7**  llm_provider CONST-036 propagation block — challenge GREEN; html/pdf regenerated.
- **W8**  anti_bluff_execution_challenge.sh BSD-sed — portable tmpfile sed (§11.4.67); isolated rc=0 proof.
- **W9**  tracked .bak/.backup residue — git-history-investigated (§11.4.124), removed + .gitignore; build 0.
- **W10**  -script latin→Cyrillic — pkg/script was never imported into unified-translator; normalizeScript wired; RED→GREEN 7 tests + 3×.
- **W12**  8 challenge scripts brittle root — git -C rev-parse --show-toplevel w/ fallback (§11.4.111); all parse:OK root:OK.
- **D2**  (wave2, 7331c54) test/unit intermittent FAIL — TestVerifier/EventEmission timing race (50ms Sleep vs goroutine-dispatched handler) + data race (proven -race). Mutex eventCollector + bounded poll. BEFORE 1/10 FAIL+race → AFTER -race -count=10 10/10 PASS.
- **W2-A**  (wave2) cmd/unified-translator coverage 3.2%→18.0%, mutation-verified; + latent dangling-pointer fix (Steps []TranslationStep→[]*TranslationStep).
- **W2-B**  (wave2) pkg/storage coverage 32.9%→35.3% (Redis cache-key helpers 0%→100%, real SQLite round-trips), mutation-verified, daemon paths honest-SKIP.
- **wave3** (18c5137): **W3** pkg/translator/llm stress/chaos + **FIXED real prod data race** in BaseTranslator cache+stats (RWMutex, -race ×3); **W2** sshworker 19.6%→39.7%; **W2** batch 77.2%→90.6%.
- **wave4** (de256dd): **sshworker progress-map leak** on ctx.Done() fixed (defer cleanup, RED 300/300→GREEN -race); **D3** pkg/translator{,/llm} test-only EventBus races fixed (asyncFlag); **D7** pkg/events TestEventBus_Subscribe race fixed; **W2** verification 64.0%→73.5%; **W2** script (already 100% stmt) +11 behavioral/digraph tests.
- **D5** (9004477) **FIXED real bug** pkg/verification parseNote: CONTENT-only note (no IMPLICATIONS) silently dropped → finalize Content; RED→GREEN, mutation-verified.
- **D6** (9004477) **FIXED** verified_factory map-order flake → order-independent ElementsMatch; 10/10.
- **wave5** (af1ef47): **W14** scripts/commit_all.sh wrapper (no-force/multi-upstream/FF-only/quiescence/explicit-paths/bg-push) + hermetic selftest 13/0 (conductor-run verified) — NOT yet wired to real remotes (line-by-line

review pending); **W2** format 0%→97.6%, markdown 73.9%→80.1%, coordination 0%→92.9% + **GENUINE nil-guard fix** in TranslateWithRetry (mutation-verified: revert→nil-panic). §11.4.147: format+coordination were rate-limit-crashed at report stage; conductor verified+adopted their work (not lost, not blindly trusted).

Session total: ~39 items resolved (+ D10 pkg/distributed FULLY -race-clean 9→0 races [4 sources incl. FallbackManager.Stop() lifecycle fix]; + D9 hardware parser testability 44.6→68.4%)

Prior: ~37 items (+ models bridge wiring digital.vasic.models into the system; + D12 REAL CORS auth-bypass vuln fixed; + D11 doc)

Prior: ~34 items (+ W15: real Postgres+Redis on-demand integration proven; PG connection-pool DoS bug found+fixed; containers StartRedis helper added)

Prior: 31 items (+ wave7: grpc 0→53.3% bufconn, api 55→62.2% httptest, security +14 adversarial tests — all real-protocol/anti-bluff, auth confirmed hardened)

Prior: 29 items (+ wave6: websocket race-fix, hardware/distributed chaos tests, D10#2 api_logger PRODUCT data-race fix)

Earlier session total: 26 items (W4-W10,W12 + D2 + W2×6[unified-translator/storage/batch/sshworker/verification/script] + W3 translator + sshworker-leak + D3 + D7 + D5 + D6). Real prod bugs fixed: translator cache race, SMTP no-deadline, dangling step-ptr, sshworker map-leak, event races (D2/D3/D7), note-content loss (D5).

OPEN workable items (queue — many parallelizable, disjoint scope)

#	Item	Type	Notes / evidence
W1	Models/ (digital.vasic.models) orphan module	RESOLVED- via-bridge	Operator directed “create bridge”: pkg/modelsbridge now CONSUMES digital.vasic.models (go.mod require+replace=>./Models), contract-tested 100%, mutation-verified (146b2b5). No longer orphan. W16 (convert to git submodule) still optional/operator-gated.
W2	Coverage → ~100% (§11.4.27); ongoing	Task	Done: unified-translator 18%, storage 35.3%, batch 90.6%, sshworker 39.7%, verification 73.5%, script(100%+depth), translator/llm(+stress/chaos). Also grpc 53.3% (bufconn), api 62.2%

#	Item	Type	Notes / evidence
			(httptest), security 84.8% +adversarial, hardware/distributed chaos. NOTE: gRPC/api-server/server/redis/postgres at 0% are integration-infra-gated (need real daemons) — W15 containers-first unlocks these; honest deferral per package.
W3	Missing test types: chaos, ddos, scaling, ui, ux, full-automation (in-module)	Task	§11.4.85 chaos/stress mandatory.
W11	Wire more docs_chain contexts (Issues/Fixed/Status once they exist)	Task	docs_chain proven operational.
W13	.gitmodules section labels still PascalCase	Task (cosmetic)	Needs per-submodule .git/modules/* gitdir move.
W14	No project commit wrapper scripts/commit_all.sh (§11.4.22)	Task	IN-FLIGHT (wave5): multi-upstream push + no-force §11.4.113 + explicit-paths spec + quiescence + background-push §11.4.88 + hermetic self-test. Review before wiring to real remotes.
W15	Containers-first infra (§11.4.76)	Feature	storage DONE: real Postgres+Redis on-demand integration via containers/pkg/brokertest (StartPostgres + new StartRedis), build-tagged; pkg/storage CRUD+chaos exercised real DBs; found+fixed PG unbounded-pool DoS (default 25/5). REMAINING: api/grpc DB+LLM <i>pipeline</i> paths need operator-gated LLM keys/SSH (can't autonomously). Pattern + helpers in place for any future slice.
W16	Convert Models/ → models submodule after W1 resolved	Task	Blocked on W1.
D1		Task	§11.4.118 discovery. They use §11.4.X cascade

#	Item	Type	Notes / evidence
	docs_chain + security CONSTITUTION.md lack literal CONST-036		scheme (not CONST- NNN); challenge doesn't check them. Governance- design decision needed — do NOT blindly inject CONST-NNN (§11.4.6).
D4	pkg/script Latin→Cyrillic morpheme-boundary mis- transliteration	Bug (open)	konjugacija→коњугација (should be конјугација); injeksija, nadživeti similar. Greedy digraph matching; real fix needs morpheme/dictionary model (high blast radius) — operator-aware decision. Pinned with KNOWN-LOSSY tests.
D8	✅ FIXED (218bfa0) pkg/ markdown bare-leading --- data-loss	Bug (fixed)	Frontmatter now must begin at first non-blank line; later --- = HR/ chapter separator. RED(1ch)→GREEN(2ch), mutation-verified. Minor →dash numbering remains pinned/unfixed (D-list, low priority).
D9	✅ FIXED (fc66c62) hardware parser testability	Task	Extracted 12 pure parsers from exec callers (behavior-preserving); 44.6→68.4%, mutation- verified.
D10	3 baseline data races (FACT, captured -race)	Bug	#2 api_logger PRODUCT race ✅ FIXED (3027da2, snapshot, mutation- verified). #1 ssh_pool + #3 performance ConnectionPool. ✅ ALL FIXED (fc66c62): ssh_pool options API, perf locked-map + count- snapshot, FallbackManager.Stop() lifecycle. pkg/distributed now 0 races.
D11	✅ FIXED (b8a24f5) CLAUDE.md mislocated CORS under pkg/security	Task	Corrected: CORS is server-layer (cmd/server corsMiddleware + internal/config).
D12	✅ FIXED (b8a24f5) CORS auth-bypass vuln	Bug(security)	cmd/server corsMiddleware reflected ARBITRARY origin + Allow-Credentials:true

#	Item	Type	Notes / evidence
			under default ["*"] → any site could make credentialed cross-origin requests. Fixed: wildcard→literal * no-creds; creds only for specific allowlist. RED→GREEN, mutation-verified. Independent review: guards sound (force-reject full-argv, FF-only via merge-base -is-ancestor, multi-upstream URL-dedup, explicit-pathspect, quiescence). FOUND+FIXED cross-platform hardening: mutation grep lacked -I → GNU grep (Linux) false-aborts on binary .pdf/.html pathspects (BSD grep doesn't; §11.4.81). Self-test 13/0. ADOPTED for normal source/doc commits. Dogfood surfaced finding #2: the §11.4.84 residue scan false-positives on files that legitimately CONTAIN marker strings (the script's own MUTATION_MARKERS def, governance docs, marker-referencing tests) — inherent to grep-based detection; §11.4.84 has no escape hatch (correct), so commit such files via manual explicit-path git. commit_all.sh is for the common case.
W14b	 DONE — reviewed + adopted	Task	

NEXT phases (priority order)

- **Phase 4** — drive W2–W8 + W10: per-type test suites to ~100% (§11.4.27), real Challenges + HelixQA bank execution with captured evidence, expand real ebook translations (all formats, whole chapters, anti-bluff §11.4/§11.4.69). Subagent-driven, ≥3 parallel streams.
- **Phase 5** — containers-first infra (W15); docs_chain context expansion (W11); full docs (guides/diagrams/SQL/templates) + enterprise Website.

- **Phase 6** — RC tag `helix_translate-1.0.0-dev-0.0.1` across all submodules + main ONLY when genuinely green (§11.4.40), operator-confirmed.

Binding constraints

Anti-bluff §11.4 (real captured evidence, no false PASS) · no-force-push §11.4.113 (merge-onto-latest-main) · multi-upstream push §2.1 · flat snake_case §11.4.28/§11.4.29 · every change independently reviewed §11.4.142 · host-safety §12 (no suspend/poweroff) · no silent component removal §11.4.122 · containers-first §11.4.76 · release prefix §11.4.151 · subagent-driven default §11.4.70 · ≥ 3 parallel streams §11.4.103.

Gotchas

- `/Volumes/T7` is **case-sensitive HFS+** (NOT the internal disk — verify on the actual volume, §11.4.6).
- No `scripts/commit_all.sh`; stage explicit paths, never `git add -A`; never `git add helix_qa` (nested third-party pointer drift).
- `PIPESTATUS` empty in this shell (zsh); use `$?` directly.
- Mutation challenges can leave `.go.backup/.bak` residue — scan + clean before commit (§11.4.84).